

Analysis

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Set Up

Packages

```
# data wrangling and visualization
library(tidyverse)

# allows file paths to work across computers
library(here)

# formatting axis labels
library(scales)

# working with dates
library(lubridate)

# combining plots
library(patchwork)
```

Data

```
# read in bird survey data
birds <- read.csv(here("data", "birds.csv"))

# read in NOAA weather data
weather <- read.csv(here("data", "NOAA_daily_summaries.csv"))
```

Preliminary Code for Analyses

Cleaning Data

```
birds_clean <- birds |>
  # convert observation date to date format
  mutate(
    observation_date = ymd(observation_date),

    # create monthly grouping variable
    month = floor_date(observation_date,
                        unit = "month")
  ) |>

  # filter to only include waterfowl observations
  filter(
    e_bird_group == "Waterfowl",

    # remove repeat observations
    repeat_observation == "No",

    # keep observations within study period
    observation_date > ymd("2019-09-30"),
    observation_date < ymd("2025-10-01")
  )
```

```
weather_clean <- weather |>

  # convert date column to date format
  mutate(
    DATE = ymd(DATE),

    # create monthly grouping variable
    month = floor_date(DATE,
                        unit = "month")
  ) |>

  # filter to study period
  filter(
    DATE > ymd("2019-09-30"),
```

```
DATE < ymd("2025-10-01")
)
```

Summaries

```
# calculate total monthly precipitation
monthly_precip <- weather_clean |>

# group observations by month
group_by(month) |>

# sum precipitation within each month
summarize(
  monthly_prctp = sum(PRCP, na.rm = TRUE),

  # remove grouping after summarizing
  .groups = "drop"
)
```

```
# calculate monthly abundance by species
monthly_waterfowl <- birds_clean |>
group_by(month, species) |>
summarize(
  abundance = sum(count, na.rm = TRUE),
  .groups = "drop"
) |>
mutate(species = factor(species, levels = c(
  "Snow Goose",
  "Ross's Goose",
  "Cackling Goose (Aleutian)",
  "Canada Goose",
  "Greater White-fronted Goose",
  "Ring-necked Duck",
  "Redhead",
  "Canvasback",
  "Ruddy Duck",
  "American Wigeon",
  "Blue-winged Teal",
  "Cinnamon Teal",
  "Northern Pintail",
```

```

    "Mallard",
    "Northern Shoveler",
    "Bufflehead",
    "Hooded Merganser",
    "Green-winged Teal",
    "Gadwall"
  )))

```

```

# calculate total monthly waterfowl abundance
total_monthly_waterfowl <- birds_clean |>

# group by month
group_by(month) |>

# sum all bird counts within each month
summarize(
  total_abundance = sum(count, na.rm = TRUE),
  .groups = "drop"
)

```

Joining Data

```

# join monthly bird abundance with monthly precipitation
bird_weather_monthly <- total_monthly_waterfowl |>

# join by month
left_join(monthly_precip,
          by = "month") |>

# arrange chronologically
arrange(month) |>

# create lagged precipitation and month-of-year variables
mutate(

  # previous month's precipitation (for figure 3)
  lagged_prctp = lag(monthly_prctp),

  # numeric month of year 1-12 (for figure 3 color scale)
  month_of_year = month(month),

```

```

# season label derived from calendar month
# (Dec/Jan/Feb = Winter, Mar/Apr/May = Spring,
# Jun/Jul/Aug = Summer, Sep/Oct/Nov = Fall)
season = case_when(
  month(month) %in% c(12, 1, 2) ~ "Winter",
  month(month) %in% c(3, 4, 5) ~ "Spring",
  month(month) %in% c(6, 7, 8) ~ "Summer",
  TRUE ~ "Fall"
),
season = factor(season, levels = c("Winter", "Spring", "Summer", "Fall"))
)

```

```

# step 1: sum abundance and precipitation within each season-year combination
seasonal_by_year <- bird_weather_monthly |>

```

```

# extract water year (Oct = start of new water year)
mutate(
  water_year = if_else(month(month) >= 10,
                        year(month) + 1,
                        year(month))
) |>

```

```

group_by(season, water_year) |>
summarize(
  # total waterfowl abundance for this season in this year
  seasonal_abundance = sum(total_abundance, na.rm = TRUE),

  # total precipitation for this season in this year
  seasonal_prctp = sum(monthly_prctp, na.rm = TRUE),

  .groups = "drop"
)

```

```

# step 2: average each season's totals across all water years
seasonal_summary <- seasonal_by_year |>

```

```

group_by(season) |>
summarize(
  avg_abundance = mean(seasonal_abundance, na.rm = TRUE),
  avg_prctp      = mean(seasonal_prctp, na.rm = TRUE),
  .groups = "drop"
)

```

Statistical Analysis Code

```
# linear regression: does previous month's precipitation
# predict total monthly waterfowl abundance?
abundance_model <- lm(total_abundance ~ lagged_prdp,
                      data = bird_weather_monthly)

summary(abundance_model)
```

Call:

```
lm(formula = total_abundance ~ lagged_prdp, data = bird_weather_monthly)
```

Residuals:

Min	1Q	Median	3Q	Max
-70.76	-40.71	-18.70	10.55	206.55

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	67.589	9.375	7.209	8.62e-10 ***
lagged_prdp	5.612	3.207	1.750	0.085 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 64.38 on 63 degrees of freedom

(1 observation deleted due to missingness)

Multiple R-squared: 0.04635, Adjusted R-squared: 0.03121

F-statistic: 3.062 on 1 and 63 DF, p-value: 0.08501

Visualizations

Figure 1: Monthly Precipitation Through Time

```
ggplot(data = monthly_precip,
       mapping = aes(x = month,
                     y = monthly_prdp)) +
```

```

# shaded area beneath line adds visual depth; line shows trend
geom_area(fill = "#4682B4", alpha = 0.25) +
geom_line(color = "#7EC8E3", linewidth = 0.8) +

# label x-axis by year only
scale_x_date(
  date_breaks = "1 year",
  date_labels = "%Y",
  expand = c(0, 0)
) +

scale_y_continuous(expand = c(0, 0)) +

labs(
  title = "Monthly Precipitation Through Time",
  x      = "Date",
  y      = "Monthly Precipitation (in)"
) +

# custom dark marsh-inspired theme applied consistently across all figures
theme_minimal(base_size = 13) +
theme(
  plot.background = element_rect(fill = "#1C2B2D", color = NA),
  panel.background = element_rect(fill = "#243336", color = NA),
  panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
  panel.grid.minor = element_blank(),
  plot.title = element_text(color = "#F0EAD6", size = 20, face = "bold",
                             margin = margin(b = 12), hjust = 0),
  axis.text = element_text(color = "#B8C9C2", size = 10),
  axis.title = element_text(color = "#D4C9A8", size = 11),
  axis.ticks = element_line(color = "#3A4F52"),
  plot.margin = margin(16, 20, 12, 16)
)

```

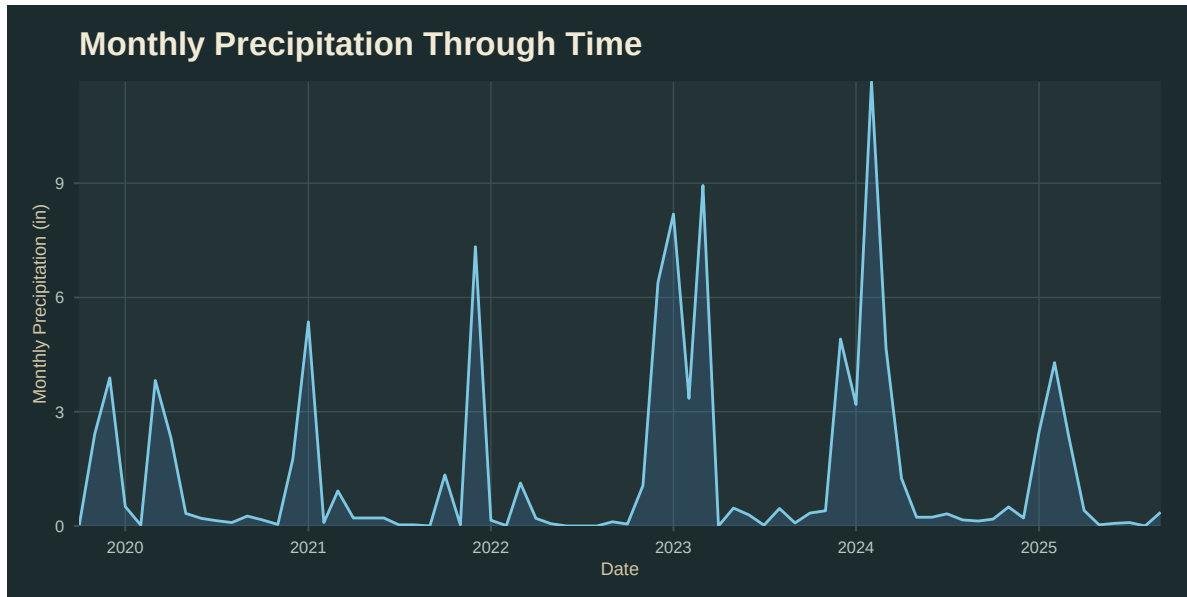


Figure 2: Monthly Waterfowl Abundance Through Time by Species

```
# manually assigned color palette for all 19 waterfowl species
waterfowl_colors <- c(
  "American Wigeon"           = "#C1440E",
  "Blue-winged Teal"         = "#1B6CA8",
  "Bufflehead"               = "#E8A838",
  "Cackling Goose (Aleutian)" = "#9DBF9E",
  "Canada Goose"             = "#6B8E23",
  "Canvasback"               = "#B22222",
  "Cinnamon Teal"            = "#A0522D",
  "Gadwall"                  = "#8FBC8F",
  "Greater White-fronted Goose" = "#D2B48C",
  "Green-winged Teal"        = "#2E8B57",
  "Hooded Merganser"         = "#20B2AA",
  "Mallard"                   = "#4682B4",
  "Northern Pintail"          = "#5F9EA0",
  "Northern Shoveler"         = "#3CB371",
  "Redhead"                   = "#8B1A1A",
  "Ring-necked Duck"         = "#6A5ACD",
  "Ross's Goose"             = "#F4E4B8",
  "Ruddy Duck"               = "#CD853F",
  "Snow Goose"               = "#B0C4DE"
```



```

)

ggplot(data = monthly_waterfowl,
       mapping = aes(x = month,
                     y = abundance,
                     fill = species)) +

# geom_col used instead of geom_area to avoid interpolation between
# sparse survey dates (geom_area would create misleading filled regions
# between months with no observations)
geom_col(color = "#1C2B2D", linewidth = 0.1) +

scale_fill_manual(values = waterfowl_colors) +

scale_x_date(
  date_breaks = "1 year",
  date_labels = "%Y",
  expand = c(0, 0)
) +

scale_y_continuous(expand = c(0, 0)) +

labs(
  title = "Monthly Waterfowl Abundance Through Time",
  x      = "Date",
  y      = "Monthly Abundance (individuals)",
  fill   = NULL
) +

theme_minimal(base_size = 13) +
theme(
  plot.background = element_rect(fill = "#1C2B2D", color = NA),
  panel.background = element_rect(fill = "#243336", color = NA),
  panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
  panel.grid.minor = element_blank(),
  plot.title       = element_text(color = "#F0EAD6", size = 20, face = "bold",
                                   margin = margin(b = 12), hjust = 0),
  plot.caption     = element_text(color = "#7A9A95", size = 8, hjust = 0,
                                   margin = margin(t = 8)),
  axis.text        = element_text(color = "#B8C9C2", size = 10),
  axis.title       = element_text(color = "#D4C9A8", size = 11),
  axis.ticks       = element_line(color = "#3A4F52"),

```

```

legend.position = "right",
legend.background = element_rect(fill = "#1C2B2D", color = NA),
legend.key       = element_rect(fill = NA, color = NA),
legend.text      = element_text(color = "#F0EAD6", size = 8.5),
legend.key.size  = unit(0.45, "cm"),
legend.spacing.y = unit(0.1, "cm"),
plot.margin = margin(16, 20, 12, 16)
)

```

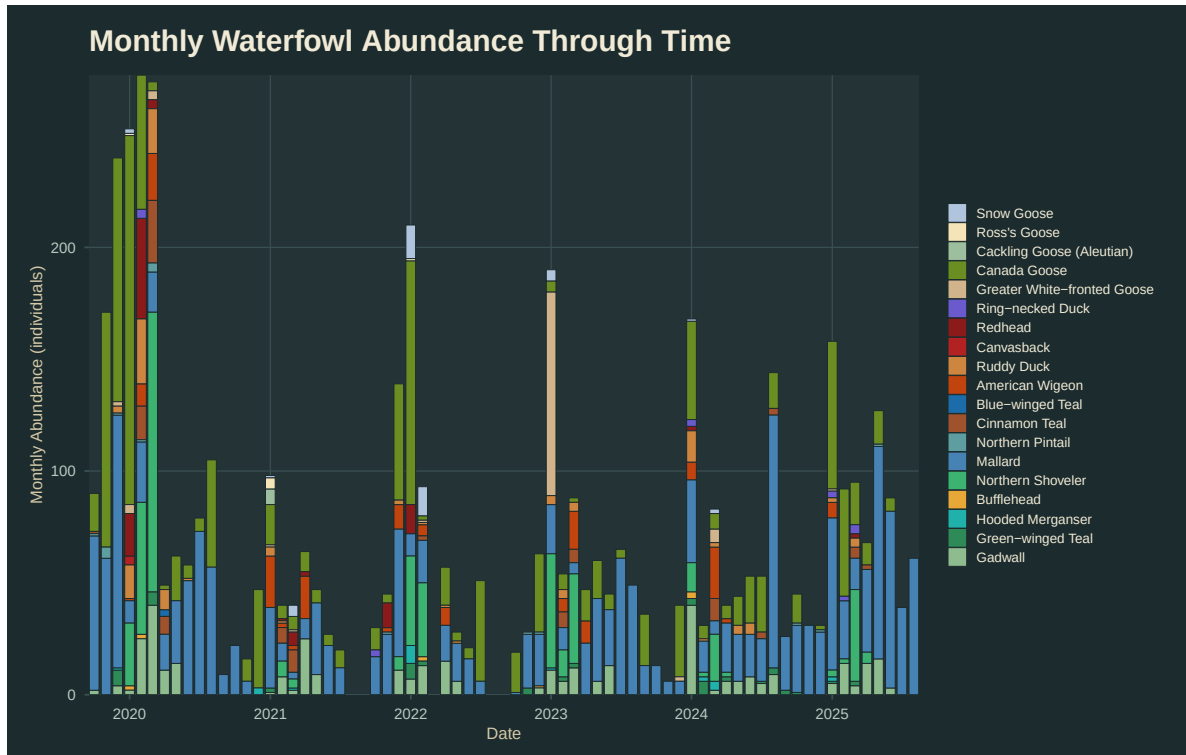


Figure 3: Lagged Precipitation vs. Total Waterfowl Abundance

```

ggplot(data = bird_weather_monthly,
  mapping = aes(x = lagged_prctp,
    y = total_abundance,
    # color points by month of year to reveal seasonal structure
    color = month_of_year)) +

geom_point(alpha = 0.85, size = 4) +

```

```

# turbo palette provides high contrast across the 1-12 month range
scale_color_viridis_c(
  name    = "Month",
  option  = "turbo",
  breaks  = c(1, 3, 5, 7, 9, 11),
  labels  = c("Jan", "Mar", "May", "Jul", "Sep", "Nov"),
  limits  = c(1, 12)
) +

labs(
  title    = "Relationship Between Precipitation and Waterfowl Abundance",
  x        = "Previous Month's Precipitation (in)",
  y        = "Total Monthly Waterfowl Abundance (individuals)",
  caption  = "Each point represents one month's waterfowl abundance plotted
against the previous month's precipitation. Points colored by month of year."
) +

theme_minimal(base_size = 13) +
theme(
  plot.background    = element_rect(fill = "#1C2B2D", color = NA),
  panel.background   = element_rect(fill = "#243336", color = NA),
  panel.grid.major    = element_line(color = "#3A4F52", linewidth = 0.4),
  panel.grid.minor    = element_blank(),
  plot.title          = element_text(color = "#F0EAD6", size = 20, face = "bold",
                                     margin = margin(b = 12), hjust = 0),
  plot.caption         = element_text(color = "#7A9A95", size = 8.5, hjust = 0,
                                     margin = margin(t = 8)),
  axis.text            = element_text(color = "#B8C9C2", size = 10),
  axis.title           = element_text(color = "#D4C9A8", size = 11),
  axis.ticks           = element_line(color = "#3A4F52"),
  legend.background    = element_rect(fill = "#1C2B2D", color = NA),
  legend.key           = element_rect(fill = NA, color = NA),
  legend.text          = element_text(color = "#F0EAD6", size = 9),
  legend.title         = element_text(color = "#D4C9A8", size = 10, face = "bold"),
  legend.key.height     = unit(1.2, "cm"),
  plot.margin          = margin(16, 20, 12, 16)
)

```

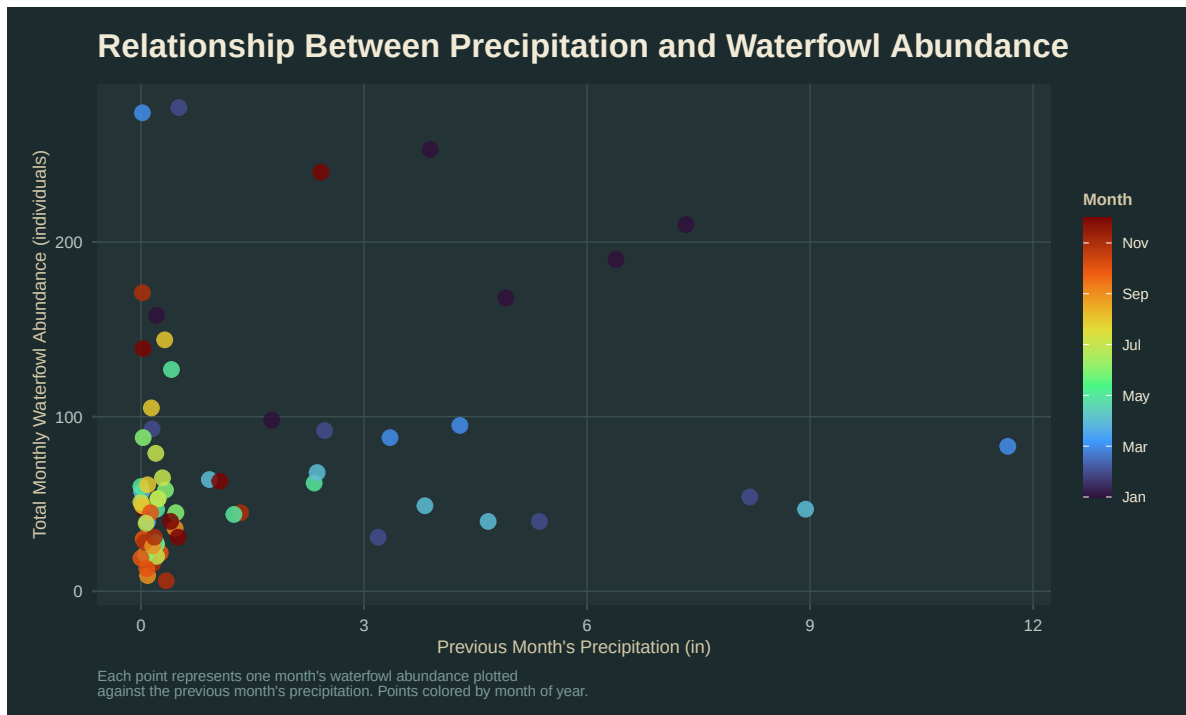


Figure 4: Average Seasonal Trends Comparing Waterfowl Abundance and Precipitation

```
# shared theme object to keep both panels visually consistent
seasonal_theme <- theme_minimal(base_size = 13) +
  theme(
    plot.background = element_rect(fill = "#1C2B2D", color = NA),
    panel.background = element_rect(fill = "#243336", color = NA),
    panel.grid.major = element_line(color = "#3A4F52", linewidth = 0.4),
    panel.grid.minor = element_blank(),
    plot.title = element_text(color = "#F0EAD6", size = 14, face = "bold",
                              margin = margin(b = 8), hjust = 0),
    axis.text = element_text(color = "#B8C9C2", size = 10),
    axis.title = element_text(color = "#D4C9A8", size = 11),
    axis.ticks = element_line(color = "#3A4F52"),
    plot.margin = margin(12, 20, 8, 16)
  )

# panel A: waterfowl abundance by season
# small jittered points = individual water-year totals; larger points + line = cross-year av
```

```

plot_abundance <- ggplot() +
  geom_point(
    data      = seasonal_by_year,
    mapping   = aes(x = season, y = seasonal_abundance, group = 1),
    color     = "#7EC8E3", alpha = 0.35, size = 2.5,
    position  = position_jitter(width = 0.08, seed = 42)
  ) +
  geom_line(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_abundance, group = 1),
    color     = "#7EC8E3", linewidth = 1.2
  ) +
  geom_point(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_abundance, group = 1),
    color     = "#7EC8E3", size = 4
  ) +
  labs(
    title = "Average Total Seasonal Waterfowl Abundance",
    x     = "Season",
    y     = "Abundance (individuals)"
  ) +
  seasonal_theme

# panel B: precipitation by season
# same structure as panel A; olive green used to distinguish precipitation from bird data
plot_precip <- ggplot() +
  geom_point(
    data      = seasonal_by_year,
    mapping   = aes(x = season, y = seasonal_prctp, group = 1),
    color     = "#6B8E23", alpha = 0.35, size = 2.5,
    position  = position_jitter(width = 0.08, seed = 42)
  ) +
  geom_line(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_prctp, group = 1),
    color     = "#6B8E23", linewidth = 1.2
  ) +
  geom_point(
    data      = seasonal_summary,
    mapping   = aes(x = season, y = avg_prctp, group = 1),
    color     = "#6B8E23", size = 4
  )

```

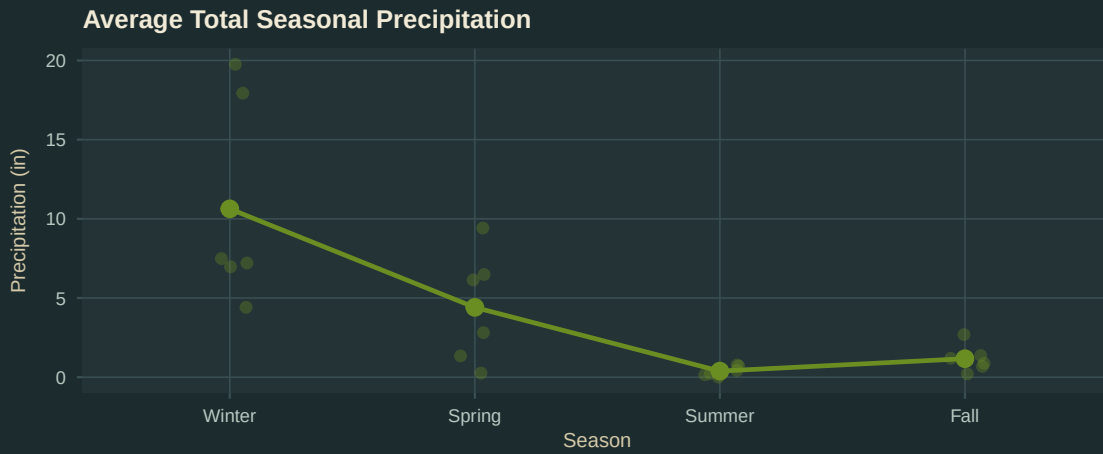
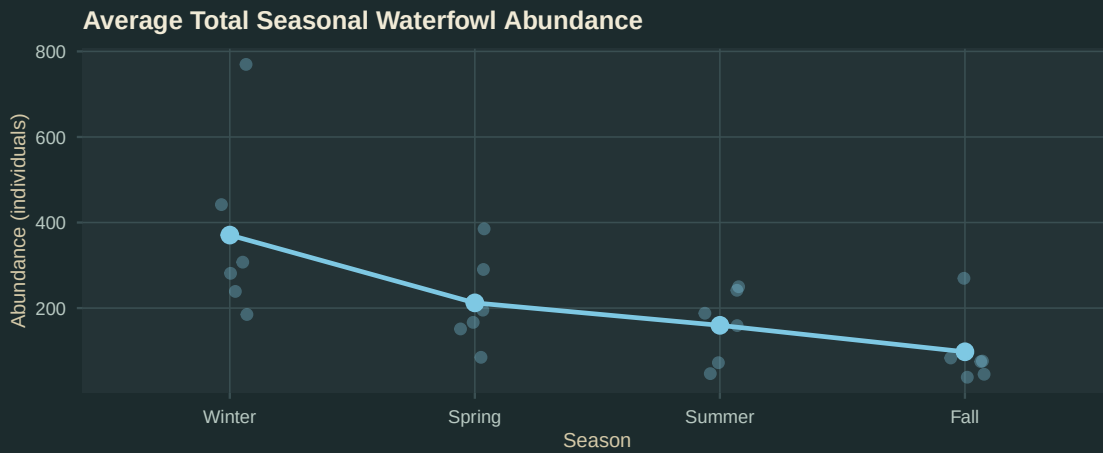
```

) +
labs(
  title = "Average Total Seasonal Precipitation",
  x      = "Season",
  y      = "Precipitation (in)"
) +
seasonal_theme

# stack panels vertically using patchwork
combined_plot <- (plot_abundance / plot_precip) +
  plot_annotation(
    title    = "Seasonal Patterns in Waterfowl Abundance and Precipitation",
    caption  = "Points show individual water-year seasonal totals; lines show averages across",
    theme    = theme(
      plot.background = element_rect(fill = "#1C2B2D", color = NA),
      plot.title       = element_text(color = "#F0EAD6", size = 18, face = "bold",
                                       hjust = 0, margin = margin(b = 10)),
      plot.caption     = element_text(color = "#7A9A95", size = 8.5, hjust = 0,
                                       margin = margin(t = 8))
    )
  )
combined_plot

```

Seasonal Patterns in Waterfowl Abundance and Precipitation



Points show individual water-year seasonal totals; lines show averages across water years 2020–2025.